

$$f(u) = f(0) + u f'(0) + \frac{u^2}{2!} f''(0) + \dots + \frac{u^n}{n!} f^{(n)}(0) + u^n \varepsilon(u)$$

$$e^u = 1 + u + \frac{u^2}{2!} + \frac{u^3}{3!} + \frac{u^4}{4!} + \dots + \frac{u^n}{n!} + u^n \varepsilon(u)$$

$$\cos u = 1 - \frac{u^2}{2!} + \frac{u^4}{4!} - \frac{u^6}{6!} + \dots + u^n \varepsilon(u)$$

$$\sin u = u - \frac{u^3}{3!} + \frac{u^5}{5!} - \frac{u^7}{7!} + \dots + u^n \varepsilon(u)$$

$$\ln(1+u) = u - \frac{u^2}{2} + \frac{u^3}{3} - \frac{u^4}{4} + \dots + u^n \varepsilon(u)$$

$$(1+u)^\alpha = 1 + \alpha u + \frac{\alpha(\alpha-1)}{2} u^2 + \frac{\alpha(\alpha-1)(\alpha-2)}{3!} u^3 + \dots + u^n \varepsilon(u)$$

$$\cosh u = 1 + \frac{u^2}{2!} + \frac{u^4}{4!} + \dots + u^n \varepsilon(u)$$

$$\sinh u = u + \frac{u^3}{3!} + \frac{u^5}{5!} + \frac{u^7}{7!} + \dots + u^n \varepsilon(u)$$

$$\alpha = -1; \frac{1}{1+u} = 1 - u + u^2 - u^3 + u^4 - u^5 + \dots$$

$$(1+u)^{1/2} = 1 + \frac{1}{2}u - \frac{1}{8}u^2 + \frac{1}{16}u^3 + \dots$$